Introduction to multiple linear regression

Benefits of multiple linear regression

> What are some benefits of including multiple predictors in my regression model?

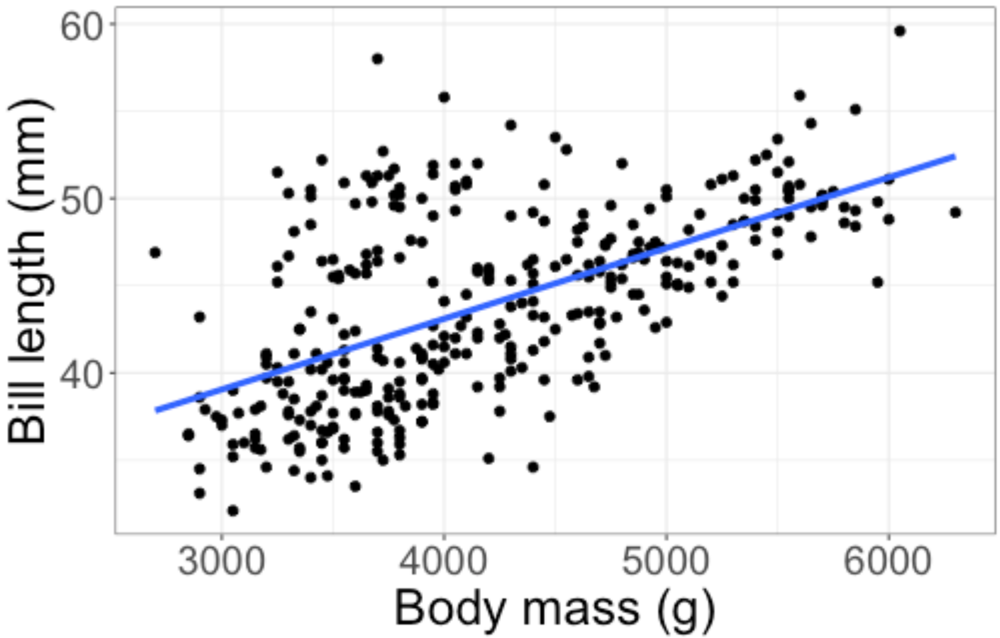
Benefits of multiple linear regression

> What are some benefits of including multiple predictors in my regression model?

Two main reasons:

- Improve the ability to predict the response
- Account for other variables that may change the relationship (e.g., penguin species)

Improving model fit

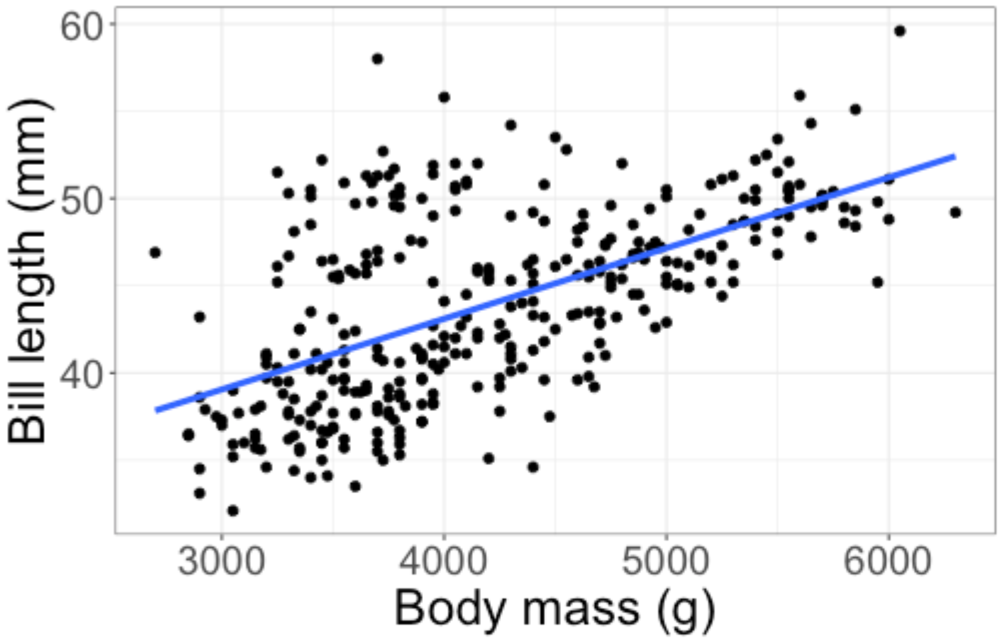


$$\widehat{\text{bill length}} = 27.15 + 0.004 \text{ body mass}$$

> How do I measure how well body mass predicts bill length?

RMSE , R^2 , R^2_{adj}

Improving model fit



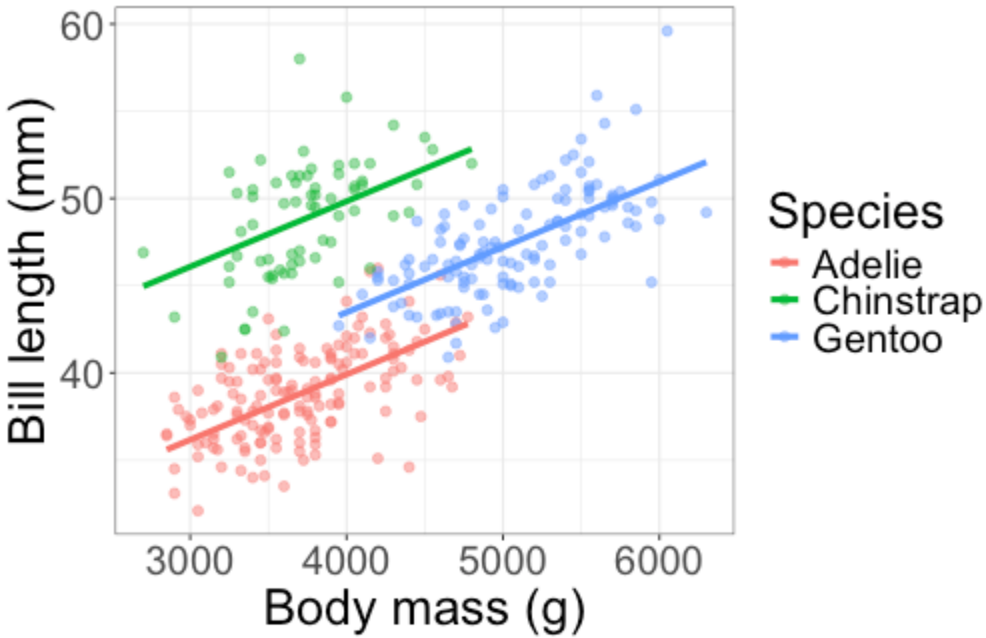
$$\widehat{\text{bill length}} = 27.15 + 0.004 \text{ body mass}$$

$R^2 = 0.348, R^2_{adj} = 0.346$

R^2_{adj}: includes penalty for # parameters

about 34.8% of variability in bill length is explained by linear regression on body mass

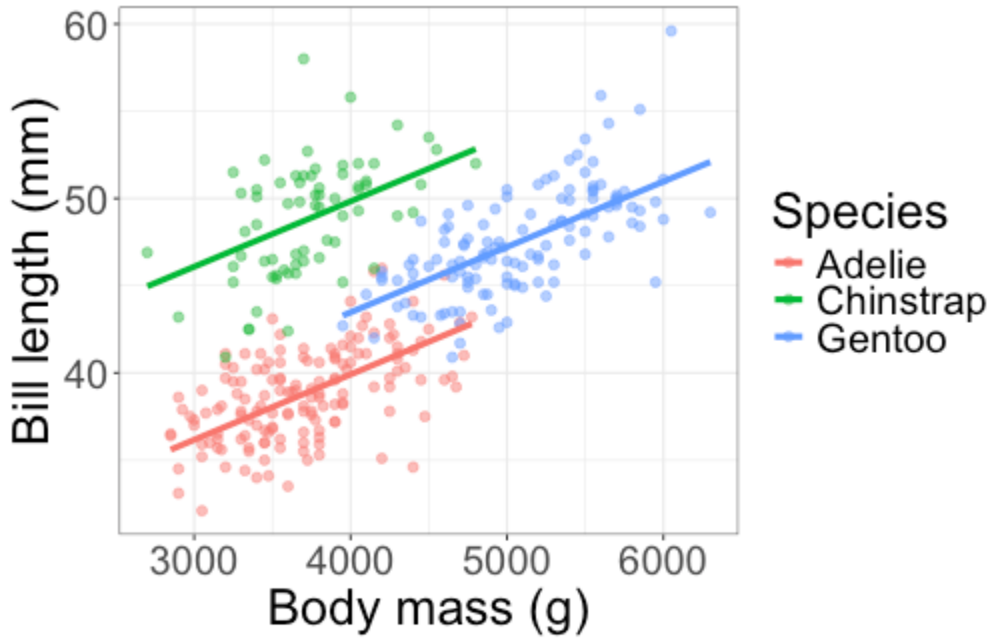
Improving model fit



But we should probably fit a line for each species...

> Do you think R^2_{adj} will increase or decrease when we add species to the model?

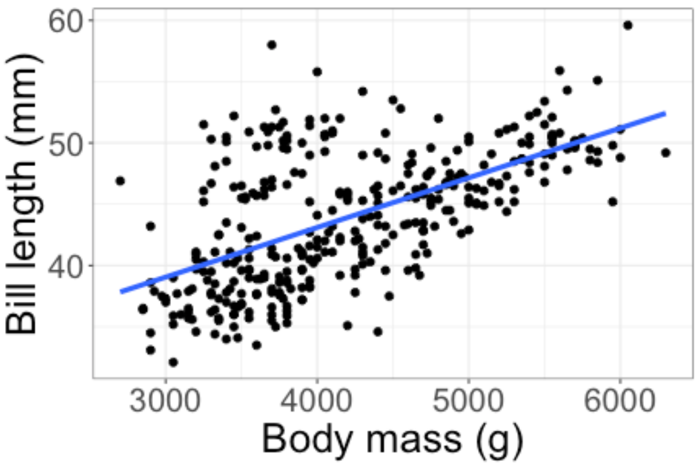
Improving model fit

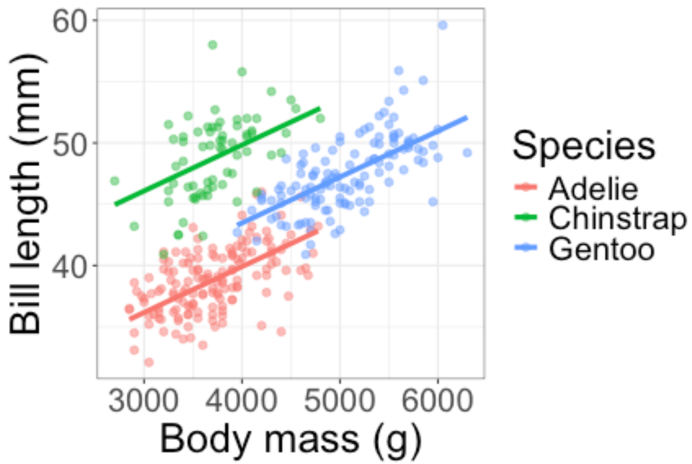


$$\widehat{\text{bill length}} = 24.92 + 9.92\ \text{IsChinstrap} + 3.56\ \text{IsGentoo} + 0.0037\ \text{body mass}$$

$R^2 = 0.806, R^2_{adj} = 0.804$

Improving model fit



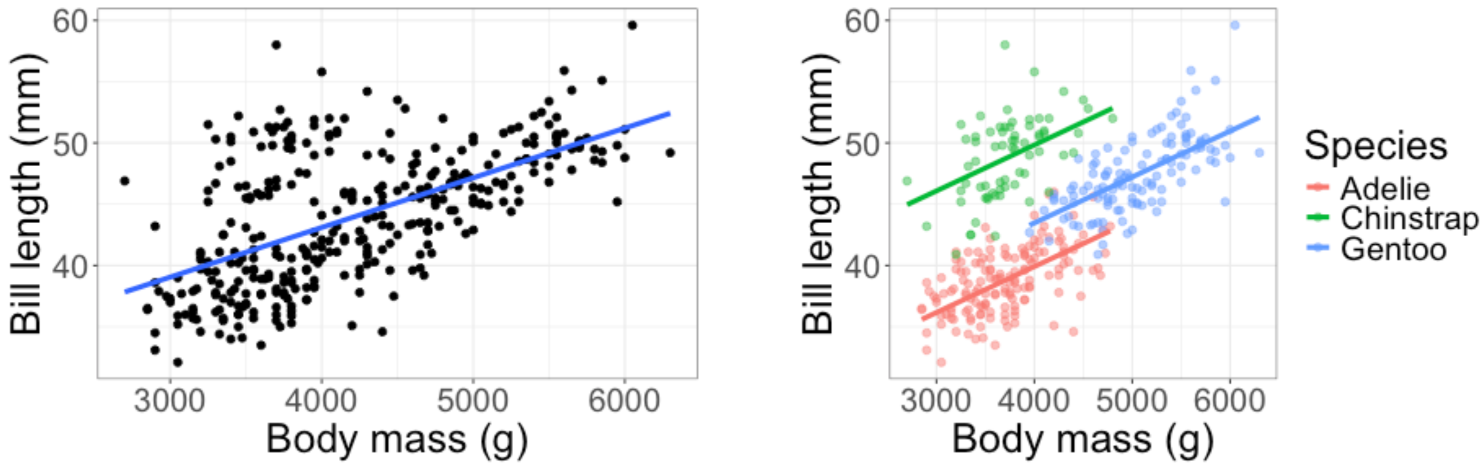


$R^2_{adj} = 0.346$

$R^2_{adj} = 0.804$

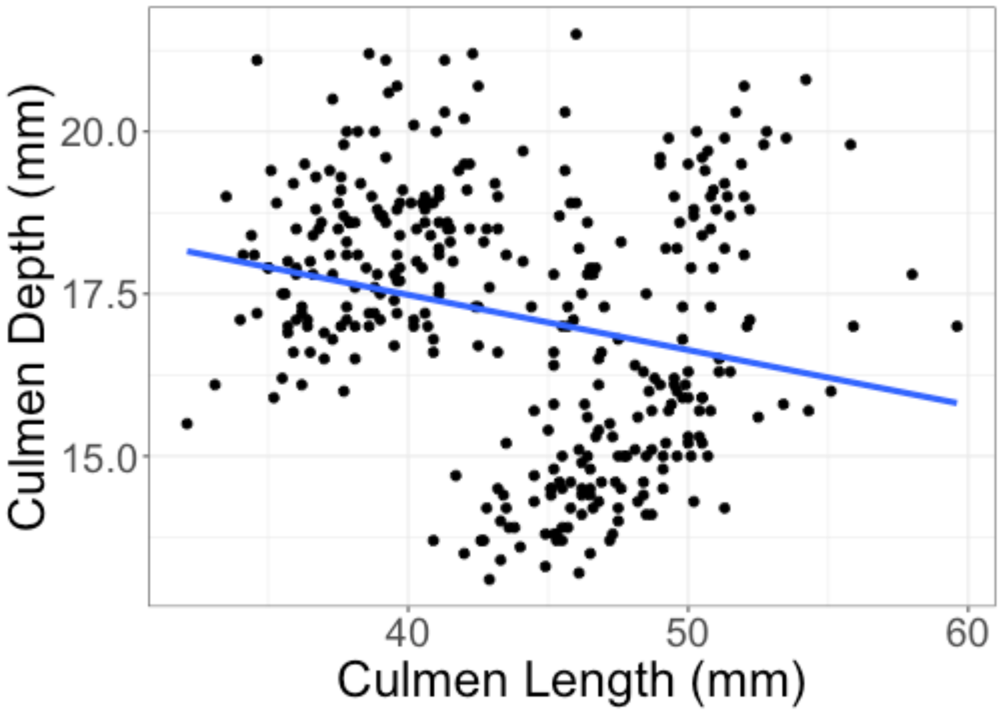
- Our ability to predict bill length increases substantially when we add Species to the model
- Make sure to compare R^2_{adj} (rather than R^2) when comparing models

Accounting for other variables



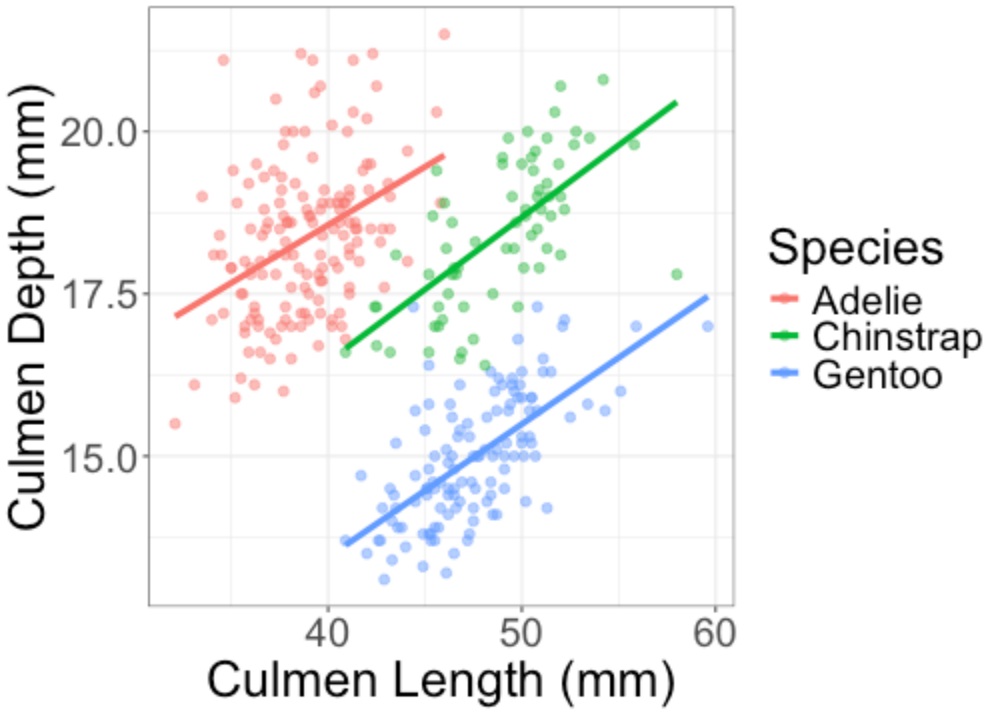
The slope looks pretty similar regardless of whether we add Species to the model. But that isn't always the case...

Accounting for other variables



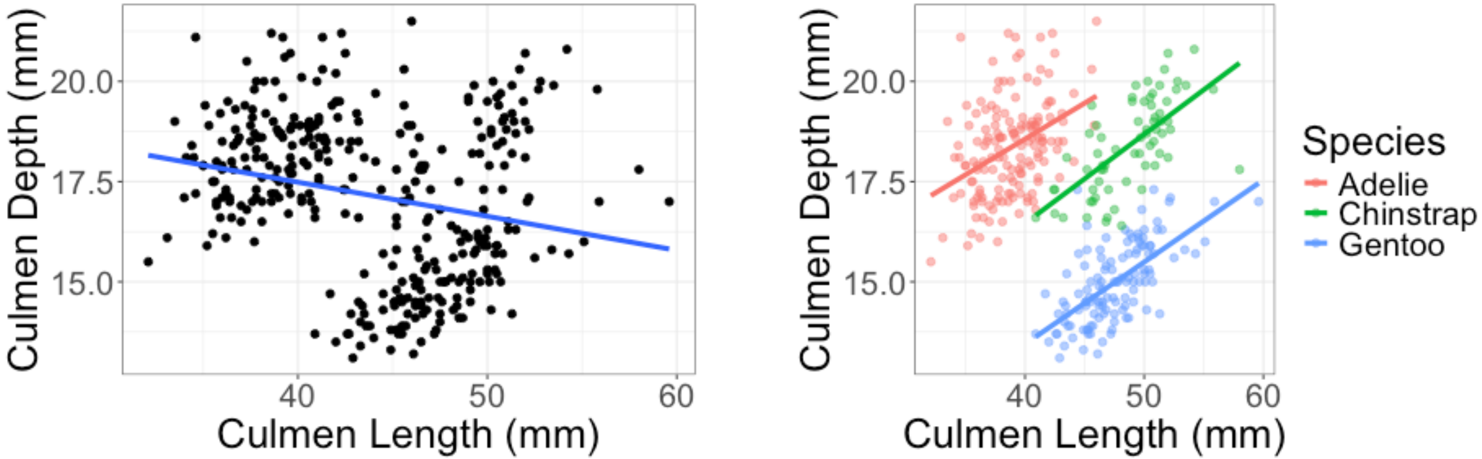
There appears to be a weak, negative relationship between culmen length and culmen depth.

Accounting for other variables



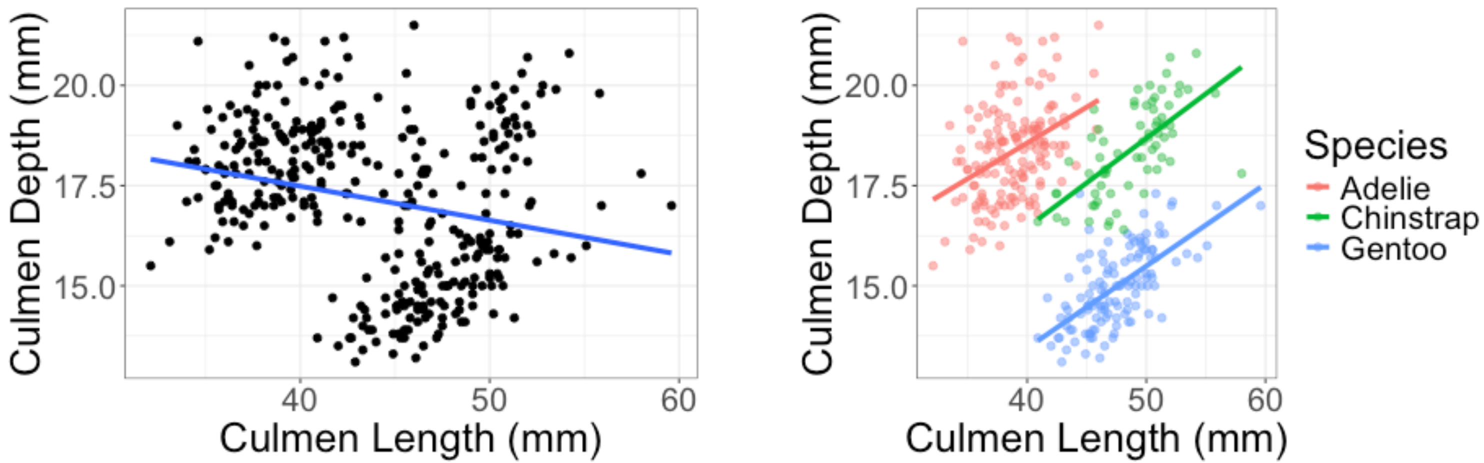
But when we include Species in the model, we see *positive* relationships between culmen length and culmen depth.

Simpson's paradox



Simpson's paradox: The apparent relationship between variables changes when you look within groups

Accounting for other variables



- The model *without* Species estimates the relationship between culmen length and culmen depth in the population of all Adelie, Chinstrap, and Gentoo penguins
- The model *with* Species estimates the relationship between culmen length and culmen depth *after accounting for species*

Two quantitative predictors

What if I want to fit a regression model with two *quantitative* predictors, body mass and flipper length?

response : bill length

> What would the population regression model look like?

bill length = $\beta_0 + \beta_1$ body mass $+ \beta_2$ flipper length $+ \varepsilon$

Two quantitative predictors

What if I want to fit a regression model with two *quantitative* predictors, body mass and flipper length?

> What would the population regression model look like?

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \beta_2 \text{flipper length} + \varepsilon$$

Interpreting coefficients

$$\widehat{\text{bill length}} = -3.44 + 0.00066 \text{ body mass} + 0.222 \text{ flipper length}$$

As with a categorical predictor, we get a separate regression line for each flipper length.

> Suppose a penguin has a flipper length of 200mm. What is the estimated regression line for the relationship between body mass and bill length?

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> Suppose a penguin has a flipper length of 200mm. What is the estimated regression line for the relationship between body mass and bill length?

The estimated regression line is

$$\begin{aligned}\widehat{\text{bill length}} &= -3.44 + 0.00066 \text{ body mass} + 0.222(200) \\ &= -3.44 + 44.4 + 0.00066 \text{ body mass} \\ &= 40.96 + 0.00066 \text{ body mass}\end{aligned}$$

Interpreting coefficients

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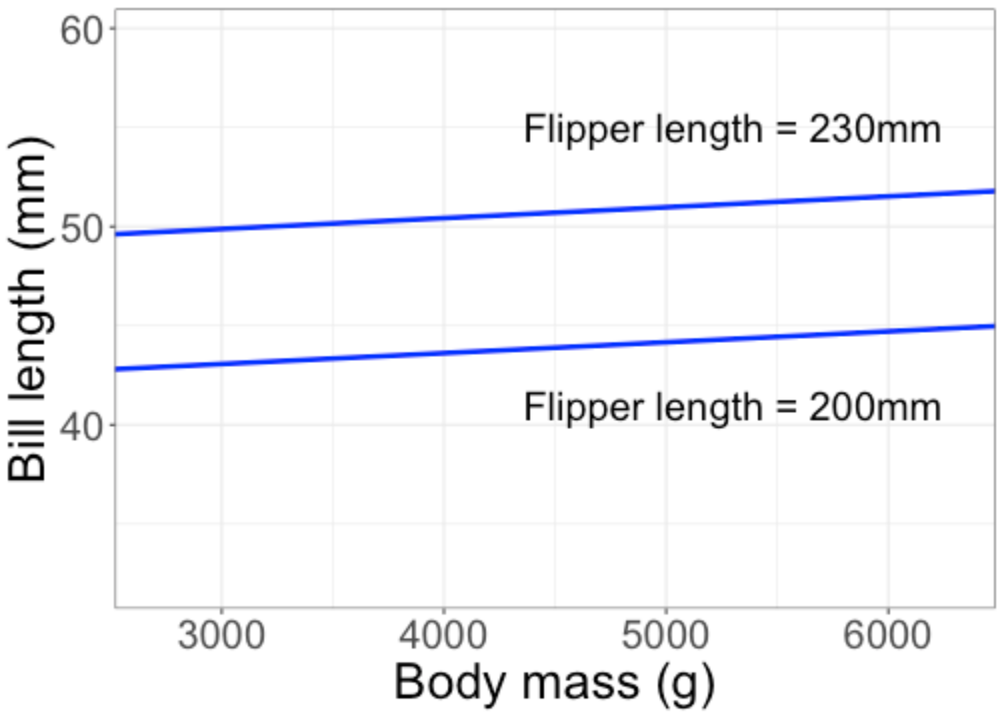

> Are longer flippers associated with longer or shorter bills?

Interpreting coefficients

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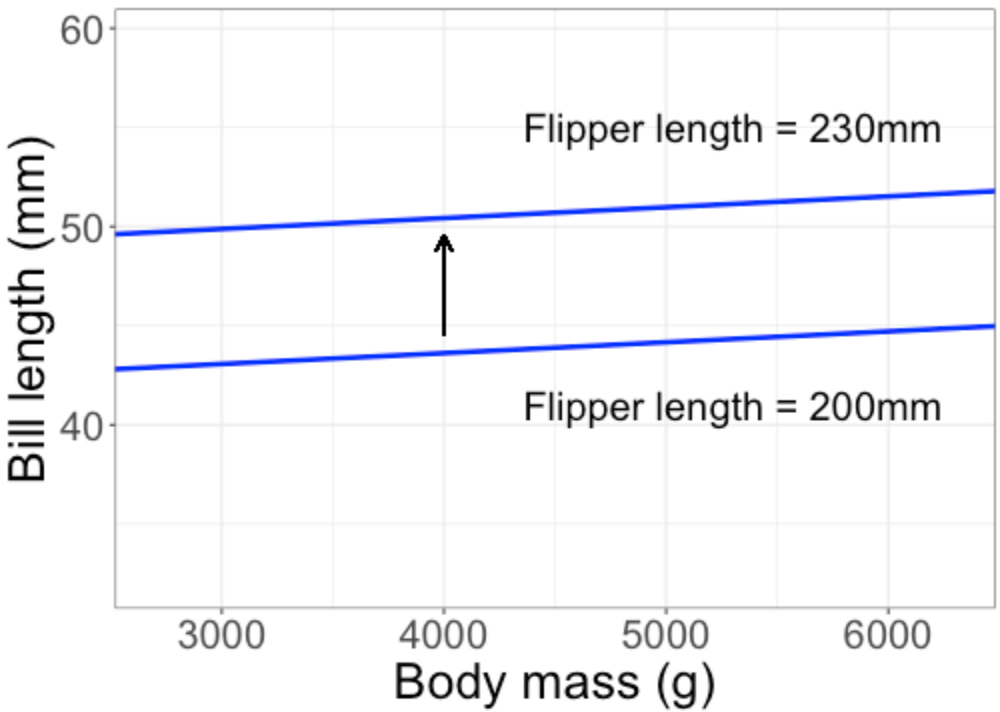
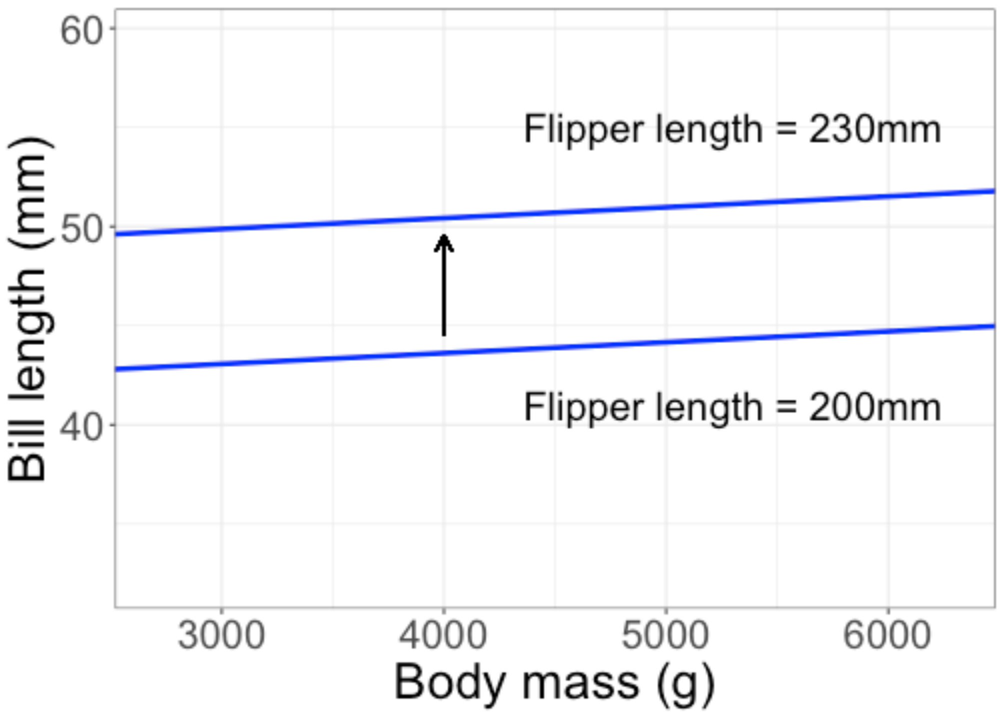


For a fixed body mass, penguins with longer flippers tend to have longer bills.

Interpreting coefficients

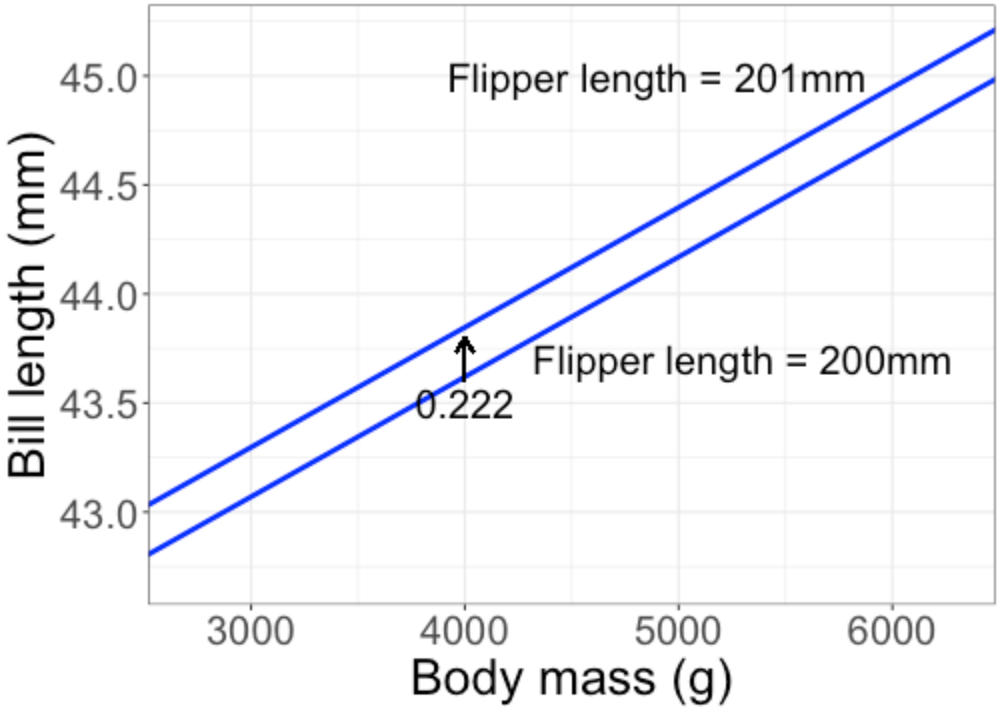
$$\widehat{\text{bill length}} = -3.44 + 0.00066\ \text{body mass} + 0.222\ \text{flipper length}$$



What does $\widehat{\beta}_2 = 0.222$ mean?

Interpreting coefficients

$$\widehat{\text{bill length}} = -3.44 + 0.00066 \text{ body mass} + 0.222 \text{ flipper length}$$



For a fixed body mass, and increase of 1mm in flipper length is associated with an increase of 0.222mm in bill length.

Class Activity

Work on the class activity (handout).

HS GPA $= 3.5$

SATV $= 600$

$\widehat{\text{College GPA}} = 0.635 + 0.498(3.5) + 0.0012(600)$

≈ 3.1

HS GPA $\quad \widehat{\text{College GPA}} = 3.5$

$3.5 = 0.635 + 0.498(3.5) + 0.0012(\text{SATV})$

$\text{SAT} = \dfrac{1.22}{0.00012} > 800$ (upper limit on SATV score)

(answer: Not possible)

Class Activity

Work on the class activity (handout).

Question: 0.0012 is much smaller than 0.498. Does that mean that SAT score is a less important/less useful predictor?

Nol.

- Scale of explanatory variables is very different
- slope is not a correlation or R^2